Research Proposal: Real-Time Monitoring and Support for Mental Health Crises

1. Title

Real-Time Monitoring and Support for Mental Health Crises: A Digital Health Intervention Utilizing Wearable Technology

2. Introduction

Mental health crises are a significant public health concern, often leading to emergency room visits, hospitalization, and long-term psychological distress. The current mental health care system is reactive, responding to crises after they occur rather than preventing them. This project proposes a digital health intervention that leverages wearable technology and mobile applications to provide real-time monitoring and support for individuals experiencing mental health crises. By tracking physiological and behavioral indicators of distress, the intervention aims to facilitate timely interventions from mental health professionals or automated support systems. This proactive approach to crisis management has the potential to reduce emergency room visits and improve mental health outcomes for at-risk individuals.

3. Problem Statement

Despite the increasing prevalence of mental health issues, access to timely and effective support remains a challenge, particularly for vulnerable populations. Traditional mental health services often lack the capacity to provide immediate assistance during crises, leading to adverse outcomes. The limitations of current interventions highlight the need for innovative solutions that can monitor individuals in real-time and provide immediate support. This research aims to address the following questions:

1. What physiological and behavioral indicators can reliably signal a mental health crisis?
2. How can wearable technology be effectively integrated into a digital health intervention for real-time monitoring?
3. What are the best practices for ensuring timely interventions from mental health professionals or automated systems?

4. Objectives

The primary objective of this research is to develop and evaluate a digital health intervention that provides real-time monitoring and support for individuals experiencing mental health crises. The specific objectives include:

1. To identify and validate physiological and behavioral indicators of mental health crises through literature review and pilot studies.
2. To design and develop a wearable technology platform that integrates with a mobile application for real-time data collection and analysis.
3. To implement a pilot study to assess the effectiveness of the intervention in reducing emergency room visits and improving mental health outcomes.
4. To evaluate user satisfaction and engagement with the digital health intervention

among vulnerable populations.

The anticipated impact of this research includes improved access to mental health support, reduced emergency room visits, and enhanced mental health outcomes for at-risk individuals.

5. Preliminary Literature Review

Recent studies have highlighted the potential of wearable technology in monitoring mental health indicators. For instance, physiological measures such as heart rate variability, skin conductance, and sleep patterns have been associated with mental health states (White Paper Brain Gaze, 2023). Additionally, behavioral indicators, including social media activity and mobile phone usage patterns, have shown promise in predicting mental health crises. However, there remains a gap in integrating these indicators into a cohesive digital health intervention that provides real-time support. This research aims to build upon existing literature by developing a comprehensive framework for real-time monitoring and intervention.

6. Methodology

The research will be conducted in three phases:

1. **Phase 1: Indicator Identification and Validation**
   - Conduct a systematic literature review to identify physiological and behavioral indicators of mental health crises.
   - Perform pilot studies to validate the identified indicators through data collection from wearable devices and mobile applications.
2. **Phase 2: Technology Development**
   - Design and develop a wearable technology platform that integrates with a mobile application for real-time data collection and analysis.
   - Collaborate with software developers and mental health professionals to ensure the platform meets clinical standards.
3. **Phase 3: Pilot Study Implementation**
   - Implement a pilot study with a sample of at-risk individuals to assess the effectiveness of the intervention.
   - Collect data on emergency room visits, mental health outcomes, and user satisfaction through surveys and interviews.

A detailed project schedule will be developed to outline the timeline for each phase, ensuring timely completion of the research objectives.

7. Expected Outcomes

The expected outcomes of this research include:

1. A validated set of physiological and behavioral indicators for real-time monitoring of mental health crises.
2. A functional digital health intervention that integrates wearable technology and mobile applications for real-time support.

3. Evidence of reduced emergency room visits and improved mental health outcomes among participants in the pilot study.
4. Insights into user engagement and satisfaction with the digital health intervention, informing future iterations and improvements.

8. Key Personnel

The research team will consist of experts in mental health, technology development, and data analysis. Key personnel include:

- **Dr. Ian McCulloh** (Principal Investigator): An expert in neuroscience and AI with extensive experience in mental health research.
- **Dr. Christophe Morin** (Co-Principal Investigator): A specialist in consumer behavior and media psychology, contributing insights into user engagement and satisfaction.

9. References

- White Paper Brain Gaze. (2023). Brain Rise Foundation. Retrieved from ProjectFiles/ResearchPapers/Papers/White Paper Brain Gaze.pdf.
- Additional references will be compiled during the literature review phase to support the research framework and methodology.

10. NOFO Alignment

This proposal aligns with the Notice of Funding Opportunity (NOFO) by testing the effectiveness of digital health platforms to improve access and continuity of care during heightened risk periods, particularly for vulnerable populations. The proactive approach to crisis management through real-time monitoring and support addresses critical gaps in current mental health services.

This proposal outlines a comprehensive plan to develop a digital health intervention for real-time monitoring and support during mental health crises, with the potential to significantly improve outcomes for at-risk individuals.